



East West University
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

COMPREHENSIVE TECHNICAL REPORT
ON
DESIGN AND IMPLEMENTATION OF A FULL-FLEDGED ENTERPRISE
NETWORK WITH MULTIPLE SUBNETS AND SERVER FARM

Course Title: Computer Networks
Course Code: CSE 405
Platform: Cisco Packet Tracer Student

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1. Introduction

The University of Scholars is a multi-campus educational institution operating critical digital infrastructure to deliver academic, research, and administrative services. In modern institutional management, reliable data communication facilitates core business processes including online admissions, course advising, results publication, eTendering, library automation, and financial workflows.

This technical report documents the end-to-end design, implementation, and evaluation of a hierarchical, enterprise-scale campus network. Modeled in Cisco Packet Tracer, the topology interconnects seven geographically separate campus networks using serial point-to-point links. The network integrates wired and wireless end-hosts, dynamic address allocation across subnets from a centralized DHCP server, centralized domain name resolution (DNS), and an official university web portal.

2. Purpose of the Network

The enterprise network satisfies the following operational goals:

- **Centralized Automated Provisioning:** Eliminate manual IP configuration across campus networks by routing DHCP broadcast requests to a centralized server in Campus 7 via DHCP Relay agents.
- **Unified Intranet Access:** Provide intranet connectivity so students and staff from any campus can access institutional web resources (<http://abc.scholarsbd.edu>).
- **Hybrid Routing Architecture:** Implement a high-performance routing topology utilizing Open Shortest Path First (OSPFv2 Area 0) across core distribution routers alongside static default and return routing for designated edge routers.
- **Multi-Tier Addressing Structure:** Segment WAN links and LAN segments across IPv4 Class A, Class B, and Class C blocks to optimize address distribution and simplify access control.
- **Hybrid Host Mediums:** Accommodate stationary computing facilities (classrooms, labs, administration) via wired FastEthernet switching while supporting mobility through Access Points on every campus.

3. Design Requirements

The technical design adheres to the following structural and protocol constraints:

Topology Layout

Seven campus routers (R1 through R7) connected via point-to-point serial connections with dedicated DCE/DTE clocking parameters (64,000 bps).

Multi-Class Address Hierarchy

- **Class A (10.0.0.0/30):** Dedicated /30 point-to-point subnets for all inter-router WAN links.
- **Class B (172.16.0.0/24):** Local Area Networks for Campuses 1, 2, and 3.
- **Class C (192.168.0.0/24):** Local Area Networks for Campuses 4, 5, 6, and the Campus 7 Server Farm.

Routing Requirements

- **Dynamic Routing:** OSPFv2 (Process ID 1, Single Area 0) configured on distribution routers R2, R3, R4, R5, and Core router R7.
- **Static Routing:** Static default routing (0.0.0.0/0) configured on edge stub routers R1 and R6, complemented with static return routes on R7.

Centralized Server Infrastructure (Campus 7)

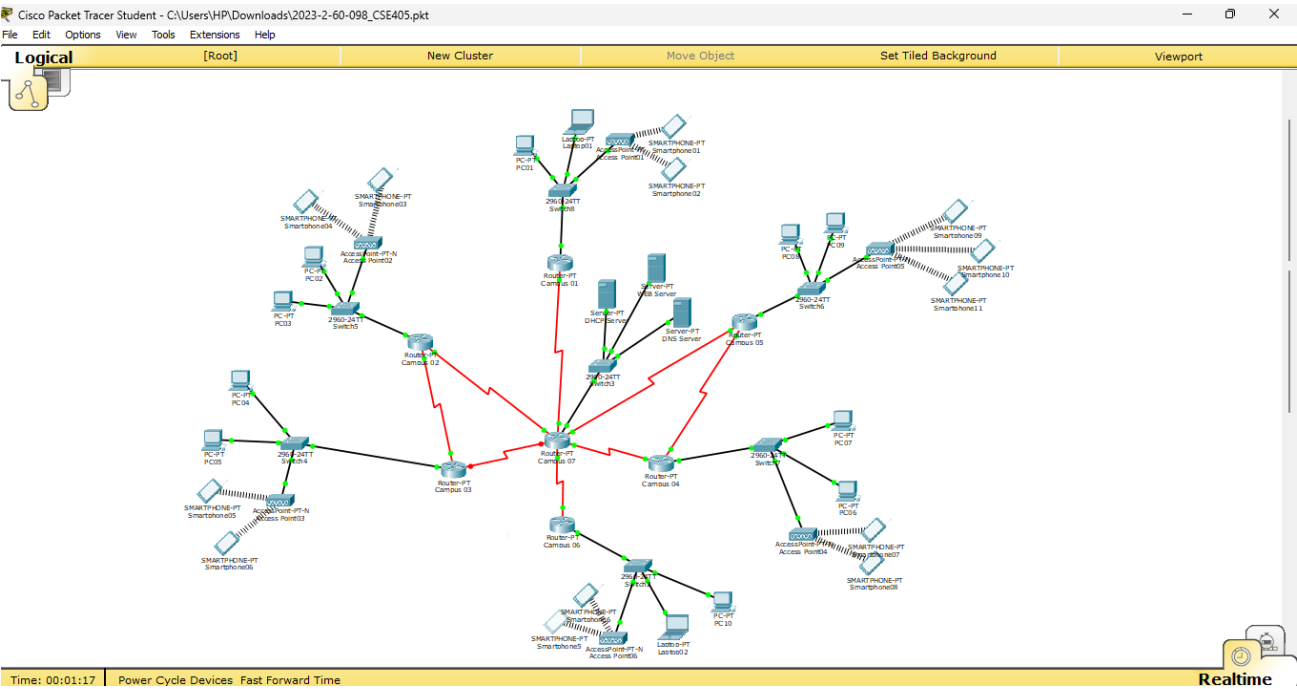
- **DHCP Server (192.168.7.10):** Single DHCP server hosting distinct pools for Campuses 1 through 6.
- **DNS Server (192.168.7.11):** Resolves fully qualified domain name abc.scholarsbd.edu to the web server.
- **Web Server (192.168.7.12):** Serves institutional web pages over HTTP/HTTPS.

LAN Composition

Each campus LAN includes at least one 2960 switch, one wireless Access Point, wired desktops, and wireless laptop. The wireless Access Point at each campus also provides the wireless network that mobile devices, such as smartphones, use to join the LAN, allowing staff and students to connect to campus resources without a wired connection.

4. Physical Network Design

The physical topology follows a modified hub-and-spoke model. Core Router R7 (Campus 7) acts as the central hub housing the Server Farm, connecting outwards to edge routers via serial links. Router R3 is reached via an inter-campus transit link through R2. Since the default router models do not ship with a serial interface, a PT-ROUTER-NM-1S module was added to each router to provide the required serial port; this module was inserted into each router before the serial WAN connections could be cabled.



5. Network Design Specifications

The table below provides comprehensive IP allocation and interface mapping for every router, WAN link, and server in the topology.

Campus / Node	Interface	IP Address	Subnet Mask	Default Gateway	Function / Connected Link
Campus 01 (R1)	Fa0/0	172.16.1.1	255.255.255.0	N/A	LAN Default Gateway (Class B)
	Ser2/0	10.0.71.2	255.255.255.252	10.0.71.1	Serial WAN Link to R7
Campus 02 (R2)	Fa0/0	172.16.2.1	255.255.255.0	N/A	LAN Default Gateway (Class B)
	Ser3/0	10.0.72.2	255.255.255.252	N/A	Serial WAN Link to R7
	Ser2/0	10.0.23.1	255.255.255.252	N/A	Inter-Campus Serial Link to R3
Campus 03 (R3)	Fa0/0	172.16.3.1	255.255.255.0	N/A	LAN Default Gateway (Class B)
	Ser2/0	10.0.23.2	255.255.255.252	10.0.23.1	Serial Link to R2 (Transit)
Campus 04 (R4)	Fa0/0	192.168.4.1	255.255.255.0	N/A	LAN Default Gateway (Class C)

Campus / Node	Interface	IP Address	Subnet Mask	Default Gateway	Function / Connected Link
	Ser2/0	10.0.74.2	255.255.255.252	N/A	Serial WAN Link to R7
Campus 05 (R5)	Fa0/0	192.168.5.1	255.255.255.0	N/A	LAN Default Gateway (Class C)
	Ser2/0	10.0.75.2	255.255.255.252	N/A	Serial WAN Link to R7
Campus 06 (R6)	Fa0/0	192.168.6.1	255.255.255.0	N/A	LAN Default Gateway (Class C)
	Ser2/0	10.0.76.2	255.255.255.252	10.0.76.1	Serial WAN Link to R7
Campus 07 (R7)	Fa0/0	192.168.7.1	255.255.255.0	N/A	Server Farm LAN Gateway
	Ser2/0	10.0.71.1	255.255.255.252	N/A	WAN to R1
	Ser3/0	10.0.72.1	255.255.255.252	N/A	WAN to R2
	Ser4/0	10.0.74.1	255.255.255.252	N/A	WAN to R4
	Ser5/0	10.0.75.1	255.255.255.252	N/A	WAN to R5
	Ser6/0	10.0.76.1	255.255.255.252	N/A	WAN to R6
DHCP Server	Fa0	192.168.7.10	255.255.255.0	192.168.7.1	Central DHCP Host
DNS Server	Fa0	192.168.7.11	255.255.255.0	192.168.7.1	Central DNS Host
Web Server	Fa0	192.168.7.12	255.255.255.0	192.168.7.1	Institutional Web Host

6. Server Configuration

6.1 DHCP Configuration

The centralized DHCP server (192.168.7.10) provides dynamic IP addressing to both wired PCs and wireless laptops across all campuses. DHCP relay agents (ip helper-address 192.168.7.10) configured on each router's LAN interface intercept broadcast requests (DHCPDISCOVER) and forward them via unicast to the server.

Pool Name	Gateway	DNS Server	Start IP Address	Subnet Mask	Max Host
Campus1	172.16.1.1	192.168.7.11	172.16.1.50	255.255.255.0	50
Campus2	172.16.2.1	192.168.7.11	172.16.2.50	255.255.255.0	50
Campus3	172.16.3.1	192.168.7.11	172.16.3.50	255.255.255.0	50
Campus4	192.168.4.1	192.168.7.11	192.168.4.50	255.255.255.0	50
Campus5	192.168.5.1	192.168.7.11	192.168.5.50	255.255.255.0	50
Campus6	192.168.6.1	192.168.7.11	192.168.6.50	255.255.255.0	50
Campus7	192.168.7.1	192.168.7.11	192.168.7.50	255.255.255.0	50

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server
campus6	192.168.6.1	192.168.7.11	192.168.6.50	255.255.255.0	50	0.0.0.0
campus5	192.168.5.1	192.168.7.11	192.168.5.50	255.255.255.0	50	0.0.0.0
campus4	192.168.4.1	192.168.7.11	192.168.4.50	255.255.255.0	50	0.0.0.0
campus3	172.16.3.1	192.168.7.11	172.16.3.50	255.255.255.0	50	0.0.0.0
campus2	172.16.2.1	192.168.7.11	172.16.2.50	255.255.255.0	50	0.0.0.0
campus1	172.16.1.1	192.168.7.11	172.16.1.50	255.255.255.0	50	0.0.0.0

DHCP Server Service Configuration:

- Open Server-PT (DHCP Server) → Services Tab → DHCP.
- Ensure Service state is set to “On”.
- Configure each pool entry as listed in the table above and click “Add” / “Save”.

6.2 DNS Configuration

The DNS server (192.168.7.11) maps the university domain name to the Web Server's IP address.

DNS Configuration Parameters:

- DNS Service: On
- Record Type: A Record
- Domain Name: abc.scholarsbd.edu
- IP Address: 192.168.7.12

DNS Resource Table:

No.	Name	Type	Resolved IP Address
1	abc.scholarsbd.edu	A Record	192.168.7.12

6.3 Web Server Configuration

The web server (192.168.7.12) runs HTTP and HTTPS daemon services to host the institutional portal.

HTTP Server Service Parameters:

- HTTP Service: On
- HTTPS Service: On
- File Name: index.html

index.html:

```
<!DOCTYPE html>
<html>
<head>
<title>University of Scholars - DNS & Web Server</title>
</head>

<body bgcolor="#EAF2F8">

<table width="100%" bgcolor="#123B63" cellpadding="20">
<tr>
<td align="center">
<font color="white" size="6">
<b>UNIVERSITY OF SCHOLARS</b>
</font>

<br><br>

<font color="#B9D9F2" size="4">
CSE405 - COMPUTER NETWORKS
</font>

<br>

<font color="white" size="3">
Fully Integrated Multi-Campus Network
</font>
</td>
</tr>
</table>

<br><br>

<table width="80%" align="center" bgcolor="white"
border="1" cellpadding="30"
bordercolor="#B8C7D1">
```



```

<tr>
<td align="center">

<font color="#123B63" size="5">
<b>UNIVERSITY WEB SERVER</b>
</font>

<br><br>

<font color="#555555" size="3">
Welcome to the official website of
University of Scholars.
</font>

<br><br><br>

<font color="#123B63" size="4">
<b>University Website Address</b>
</font>

<br><br>

<font color="#196F3D" size="5">
<b>http://abc.scholarsbd.edu</b>
</font>

<br><br>

<font color="#555555" size="3">
A single DNS server is used to locate
the University Web Server.
</font>

<br><br>

<font color="#555555" size="3">
Users from any campus can access the
University website using the above domain name.
</font>

</td>
</tr>

</table>

<br><br>

<table width="80%" align="center"
bgcolor="white"
border="1"
cellpadding="15"
bordercolor="#B8C7D1">

<tr bgcolor="#123B63">
<td align="center">
<font color="white">
<b>NETWORK SERVICE</b>
</font>
</td>

<td align="center">
<font color="white">
<b>ROLE</b>
</font>
</td>

<td align="center">
<font color="white">
<b>STATUS</b>
</font>
</td>
</tr>

<tr>

```

```

<td align="center">DNS Server</td>

<td align="center">
Locates Web Server
</td>

<td align="center">
<font color="green">
<b>ONLINE</b>
</font>
</td>
</tr>

<tr>
<td align="center">Web Server</td>

<td align="center">
Hosts University Website
</td>

<td align="center">
<font color="green">
<b>ONLINE</b>
</font>
</td>
</tr>

<tr>
<td align="center">Multi-Campus Network</td>

<td align="center">
Provides Campus Connectivity
</td>

<td align="center">
<font color="green">
<b>CONNECTED</b>
</font>
</td>
</tr>

</table>

<br><br>

<table width="100%" bgcolor="#123B63" cellpadding="15">
<tr>
<td align="center">

<font color="white" size="2">
<b>UNIVERSITY OF SCHOLARS</b>
<br><br>
CSE405 - COMPUTER NETWORKS
<br>
Single DNS Server Based Web Access
<br><br>
Website:
<b>http://abc.scholarsbd.edu</b>
<br><br>

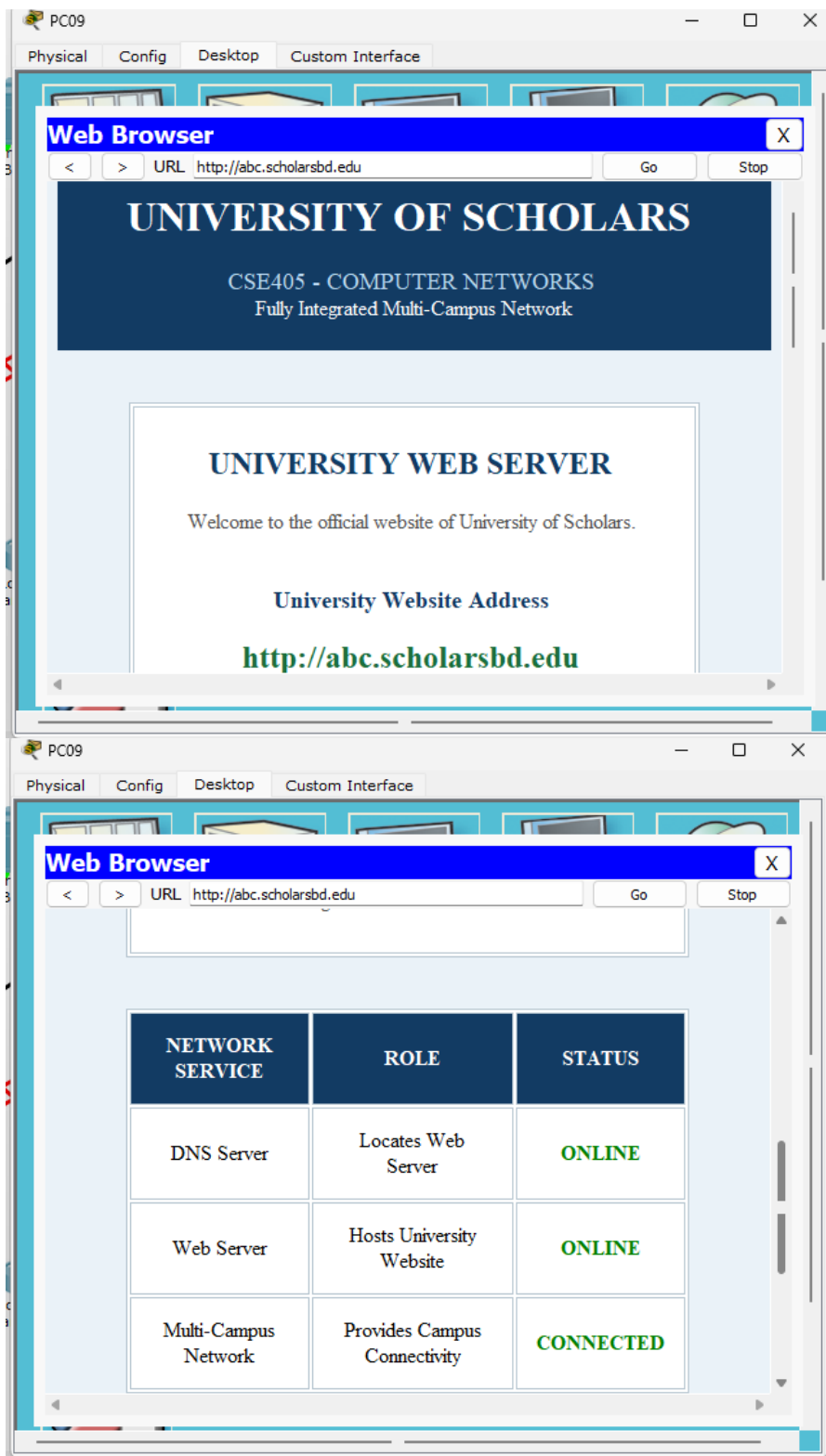
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</font>

</td>
</tr>
</table>

</body>
</html>

```



7. Router Configuration

7.1 Campus 01 (R1 - Static Routing)

```
Router> enable
Router# configure terminal
hostname R1
interface FastEthernet0/0
  ip address 172.16.1.1 255.255.255.0
  ip helper-address 192.168.7.10
  no shutdown
exit
interface Serial2/0
  ip address 10.0.71.2 255.255.255.252
  no shutdown
exit
ip route 0.0.0.0 0.0.0.0 10.0.71.1
end
write memory
```

7.2 Campus 02 (R2 - Dynamic OSPF & Transit Link)

```
Router> enable
Router# configure terminal
hostname R2
interface FastEthernet0/0
  ip address 172.16.2.1 255.255.255.0
  ip helper-address 192.168.7.10
  no shutdown
exit
interface Serial3/0
  ip address 10.0.72.2 255.255.255.252
  no shutdown
exit
interface Serial2/0
  ip address 10.0.23.1 255.255.255.252
  clock rate 64000
  no shutdown
exit
router ospf 1
  router-id 2.2.2.2
  network 172.16.2.0 0.0.0.255 area 0
  network 10.0.72.0 0.0.0.3 area 0
  network 10.0.23.0 0.0.0.3 area 0
exit
ip route 0.0.0.0 0.0.0.0 10.0.72.1
ip route 172.16.3.0 255.255.255.0 10.0.23.2
end
write memory
```

7.3 Campus 03 (R3 - Dynamic OSPF)

```
Router> enable
Router# configure terminal
hostname R3
interface FastEthernet0/0
 ip address 172.16.3.1 255.255.255.0
 ip helper-address 192.168.7.10
 no shutdown
exit
interface Serial2/0
 ip address 10.0.23.2 255.255.255.252
 no shutdown
exit
router ospf 1
 router-id 3.3.3.3
 network 172.16.3.0 0.0.0.255 area 0
 network 10.0.23.0 0.0.0.3 area 0
exit
ip route 0.0.0.0 0.0.0.0 10.0.23.1
end
write memory
```

7.4 Campus 04 (R4 - Dynamic OSPF)

```
Router> enable
Router# configure terminal
hostname R4
interface FastEthernet0/0
 ip address 192.168.4.1 255.255.255.0
 ip helper-address 192.168.7.10
 no shutdown
exit
interface Serial2/0
 ip address 10.0.74.2 255.255.255.252
 no shutdown
exit
router ospf 1
 router-id 4.4.4.4
 network 192.168.4.0 0.0.0.255 area 0
 network 10.0.74.0 0.0.0.3 area 0
exit
ip route 0.0.0.0 0.0.0.0 10.0.74.1
end
write memory
```

7.5 Campus 05 (R5 - Dynamic OSPF)

```
Router> enable
Router# configure terminal
hostname R5
interface FastEthernet0/0
 ip address 192.168.5.1 255.255.255.0
 ip helper-address 192.168.7.10
 no shutdown
exit
interface Serial2/0
 ip address 10.0.75.2 255.255.255.252
 no shutdown
exit
router ospf 1
 router-id 5.5.5.5
 network 192.168.5.0 0.0.0.255 area 0
 network 10.0.75.0 0.0.0.3 area 0
exit
ip route 0.0.0.0 0.0.0.0 10.0.75.1
end
write memory
```

7.6 Campus 06 (R6 - Static Routing)

```
Router> enable
Router# configure terminal
hostname R6
interface FastEthernet0/0
 ip address 192.168.6.1 255.255.255.0
 ip helper-address 192.168.7.10
 no shutdown
exit
interface Serial2/0
 ip address 10.0.76.2 255.255.255.252
 no shutdown
exit
ip route 0.0.0.0 0.0.0.0 10.0.76.1
end
write memory
```

7.7 Campus 07 (R7 - Core Router & Server Farm Gateway)

```
Router> enable
Router# configure terminal
hostname R7
interface FastEthernet0/0
 ip address 192.168.7.1 255.255.255.0
 no shutdown
exit
interface Serial2/0
 ip address 10.0.71.1 255.255.255.252
 clock rate 64000
 no shutdown
exit
interface Serial3/0
 ip address 10.0.72.1 255.255.255.252
 clock rate 64000
 no shutdown
exit
interface Serial4/0
 ip address 10.0.74.1 255.255.255.252
 clock rate 64000
 no shutdown
exit
interface Serial5/0
 ip address 10.0.75.1 255.255.255.252
 clock rate 64000
 no shutdown
exit
interface Serial6/0
 ip address 10.0.76.1 255.255.255.252
 clock rate 64000
 no shutdown
exit
router ospf 1
 router-id 7.7.7.7
 network 192.168.7.0 0.0.0.255 area 0
 network 10.0.72.0 0.0.0.3 area 0
 network 10.0.74.0 0.0.0.3 area 0
 network 10.0.75.0 0.0.0.3 area 0
exit
ip route 172.16.1.0 255.255.255.0 10.0.71.2
ip route 192.168.6.0 255.255.255.0 10.0.76.2
ip route 172.16.3.0 255.255.255.0 10.0.72.2
ip route 10.0.23.0 255.255.255.252 10.0.72.2
end
write memory
```

8. Accessing the Website via Smartphone and PC and laptop

The figure displays six screenshots of a network simulation environment, showing how a website is accessed from different devices. Each device has a 'Web Browser' window open to the URL <http://abc.scholarsbd.edu>.

Smartphone09 (Top Left): Displays the 'UNIVERSITY OF SCHOLARS' header, 'CSE405 - COMPUTER NETWORKS' subtitle, and 'Fully Integrated Multi-Campus Network' text. Below is the 'UNIVERSITY WEB SERVER' section with a welcome message and a placeholder for the 'University Website Address'.

Smartphone09 (Top Right): Displays a table of network services and their status.

NETWORK SERVICE	ROLE	STATUS
DNS Server	Locates Web Server	ONLINE
Web Server	Hosts University Website	ONLINE
Multi-Campus Network	Provides Campus Connectivity	CONNECTED

Laptop01 (Middle Left): Displays the 'UNIVERSITY WEB SERVER' section with a welcome message and the 'University Website Address' as <http://abc.scholarsbd.edu>. It also includes explanatory text about the DNS server and user access.

Laptop01 (Middle Right): Displays the same table of network services and their status as seen on Smartphone09.

PC02 (Bottom Left): Displays the 'UNIVERSITY OF SCHOLARS' header, 'CSE405 - COMPUTER NETWORKS' subtitle, and 'Fully Integrated Multi-Campus Network' text. Below is the 'UNIVERSITY WEB SERVER' section with a welcome message and the 'University Website Address' as <http://abc.scholarsbd.edu>.

PC02 (Bottom Right): Displays the table of network services and their status, and a footer section with the 'UNIVERSITY OF SCHOLARS' header, 'CSE405 - COMPUTER NETWORKS' subtitle, 'Single DNS Server Based Web Access' text, and the 'Website: <http://abc.scholarsbd.edu>'.

9. Connection Between Devices Across All Campuses

PC-PT

PDU List Window

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC07	PC09	ICMP		0.000	N	0	(edit)	(delete)
	Successful	PC09	Laptop01	ICMP		0.000	N	1	(edit)	(delete)
	Successful	PC01	PC02	ICMP		0.000	N	2	(edit)	(delete)
	Successful	PC03	PC04	ICMP		0.000	N	3	(edit)	(delete)
	Successful	Smart...	PC10	ICMP		0.000	N	4	(edit)	(delete)
	Successful	PC10	PC06	ICMP		0.000	N	5	(edit)	(delete)
	Successful	Smart...	Smartphone01	ICMP		0.000	N	6	(edit)	(delete)
	Successful	Smart...	Laptop01	ICMP		0.000	N	7	(edit)	(delete)
	Successful	PC07	PC04	ICMP		0.000	N	8	(edit)	(delete)

Time: 00:09:02 Power Cycle Devices Fast Forward Time Realtime

Scenario 0

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC04	Smartphone6	ICMP		0.000	N	0	(edit)	(delete)
	Successful	PC09	PC08	ICMP		0.000	N	1	(edit)	(delete)
	Successful	PC06	PC07	ICMP		0.000	N	2	(edit)	(delete)
	Successful	PC05	Smartphone05	ICMP		0.000	N	3	(edit)	(delete)
	Successful	PC02	Smartphone04	ICMP		0.000	N	4	(edit)	(delete)
	Successful	PC09	Smartphone09	ICMP		0.000	N	5	(edit)	(delete)
	Successful	Laptop...	PC10	ICMP		0.000	N	6	(edit)	(delete)
	Successful	Smart...	Smartphone02	ICMP		0.000	N	7	(edit)	(delete)

Time: 00:12:04 Power Cycle Devices Fast Forward Time Realtime

10. Limitations

While the network fulfills all functional and academic requirements, certain structural limitations exist in this model:

- **Single Point of Failure (SPOF):** Consolidating DHCP, DNS, and HTTP services into a single physical server subnet in Campus 7 creates vulnerability. If Core Router R7 or the Server Farm switch fails, all network provisioning and intranet services cease across the institution.
- **Transit Link Dependency:** Campus 3 relies on Campus 2 (R2) as a transit path. Disruption along the 10.0.23.0/30 link isolates Campus 3 from the rest of the university.
- **Lack of Redundant WAN Mesh:** Edge routers R1, R4, R5, and R6 connect to R7 via single serial links with no secondary or backup paths.
- **IPv4 Address Exhaustion in Real-World Deployment:** Utilizing public Class A and Class B subnets internally is standard in simulation environments, but production deployments require Private Address Space (RFC 1918) combined with Network Address Translation (NAT/PAT) at edge routers.
- **Absence of Access Control Lists (ACLs):** Inter-campus traffic currently flows unrestricted without firewall or ACL filtering between administrative, student, and server zones.
- **Direct Serial Link Inactivity & Transit Dependency (Campus 03 ↔ Campus 07):** The direct physical serial link between R3 and R7 remained inactive (down/down), requiring Campus 03 to route all intranet and DHCP traffic via the intermediate transit hop at Campus 02 (R_2) over subnet 10.0.23.0/30. While full inter-campus reachability and DNS/Web browsing function properly through R_2 , activating and configuring OSPF over the direct $R_3 \leftrightarrow R_7$ link (10.0.73.0/30) would provide link redundancy and eliminate single-transit dependency.

11. Conclusion

The University of Scholars multi-campus network project was successfully designed, configured, and validated. By combining dynamic OSPF routing on core distribution routers with static routing on edge routers, the system achieves efficient routing and complete inter-subnet reachability. Centralized DHCP Relay agents allow a single server to dynamically lease IP configurations across all seven LAN segments, while DNS name resolution routes user web traffic to the official university portal at <http://abc.scholarsbd.edu>. The network design satisfies all technical requirements, providing an interconnected foundation for institutional communications.